

Comparing data sets



1 The number of goals scored by Team 1 and Team 2 in a football tournament are recorded in the following table:

- a Find the total number of goals scored by both teams in Match C.
- b Find the total number of goals scored by Team 1 across all the matches.
- c Calculate the mean number of goals scored by:
 - i Team 1
 - ii Team 2

Match	Team 1	Team 2
A	2	3
B	4	2
C	5	2
D	3	5
E	3	4

2 The table shows the scores of Student A and Student B in 5 separate tests:

- a Find the mean of the scores of:
 - i Student A
 - ii Student B
- b What is the mean of the combined scores of the two students?
- c What is the highest score overall and which student obtained that score?
- d What is the lowest score overall and which student obtained that score?

Test	Student A	Student B
1	79	77
2	74	83
3	86	72
4	89	94
5	98	97

3 The salaries of men and women working the same job at the same company are given below:

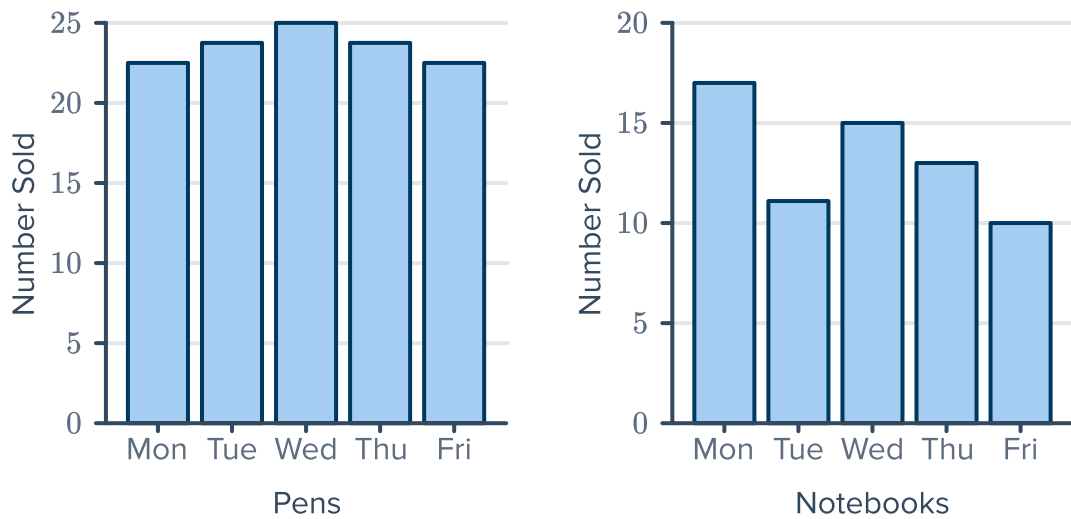
Men	80 000	80 000	75 000	80 000	75 000	70 000	80 000
Women	70 000	70 000	75 000	70 000	70 000	80 000	75 000

- a Copy and complete the table, giving your answers correct to two decimal place where appropriate:

	Men	Women
Mean		
Median		
Mode		
Range		

- b Which gender seems to be paid better? Justify your answer.

- 4 A stationery shop kept a weekly record of  number of pens and notebooks sold:



- a Find the number of pens sold on Tuesday.
- b Find the number of notebooks sold on Friday.
- c Find the total number of notebooks sold during the week.
- d Find the total number of pens sold during the week
- e Find the percentage of pens sold on Thursday, rounding your answer to two decimal places.
- f Calculate the mean number of notebooks sold per day. Round your answer to one decimal place.
- g Calculate the mean number of pens sold per day. Round your answer to one decimal place.
- h Which is the better selling product? Justify your answer.

- 5 The runs scored by each player in a cricket match are displayed below:

- Team A: 63, 0, 81, 22, 39, 66, 9, 18, 3, 3, 5
- Team B: 21, 26, 18, 69, 62, 26, 5, 28, 2, 1, 2

- a Find the mean runs of the following teams, rounded to one decimal place:
 - i Team A
 - ii Team B
- b Which team scored more runs?

6 In a standardised test out of 100, the students in two classes got the results displayed below:

- Class A: 70, 64, 94, 78, 79, 61, 69, 93, 64, 79, 64, 71, 82, 68, 66
- Class B: 86, 68, 51, 80, 63, 87, 79, 89, 62, 79, 84, 77, 67, 80, 75

a Find the median mark for:

i Class A

ii Class B

b Using the medians, which class did better on the test?

7 The monthly rainfall for two cities across a year is displayed below:

- City A: 122, 120, 121, 67, 79, 24, 8, 46, 27, 67, 80, 128
- City B: 51, 66, 33, 78, 79, 92, 78, 99, 41, 60, 26, 56

a Find the range in monthly rainfall for:

i City A

ii City B

b Which city had more consistent rainfall throughout the year?

8 Marge grows two different types of bean plants. She records the number of beans that she picks from each plant for 10 days. Her records are shown below:

- Plant A: 10, 4, 4, 5, 7, 10, 3, 3, 9, 10
- Plant B: 8, 7, 5, 5, 9, 7, 8, 7, 5, 6

a What is the mean number of beans picked per day for Plant A? Round your answer to one decimal place.

b What is the mean number of beans picked per day for Plant B? Round your answer to one decimal place.

c What is the range for Plant A?

d What is the range for Plant B?

e Which plant produces more beans on average?

f Which plant has a more consistent yield of beans?

9 The weight (in kilograms) of a group of men and women were recorded and presented in a stem and leaf plot as shown:

Women		Men
7 6 3	5	
8 6 3 1 1	6	2 3 3 8 9
3 1	7	1 2 4 8
	8	2

a Find the mean weight of the group of men.

b Find the mean weight of the group of

- c Which group is heavier?



Key: $8|3 = 83$

10 The stem and leaf plot shows the batting scores of two cricket teams, India and Sri Lanka:

- a Find the highest score in:

- i India ii Sri Lanka

- b Find the mean score of:

- i India ii Sri Lanka

- c Calculate the combined mean of the two teams.

India		Sri Lanka
9 6	3	0 2 5 6 9
9 7 3 1 1	4	2 7
7 2 0	5	6 8
	6	3

Key: $8|3 = 83$

11 The stem and leaf plot shows the test scores of two classes, A and B:

- a Find the highest score in:

- i Class A ii Class B

- b Find the mean score of:

- i Class A ii Class B

- c Calculate the combined mean of the two classes, correct to two decimal places.

Class A		Class B
7 3 2	6	2 7
9 8 7 1 0	7	1 3 6 7
3 0	8	1 2 4 6
	9	1 8

Key: $8|3 = 83$

12 The stem and leaf plot shows the number of books read in a year by a random sample of university and high school students:

- a Interpret the lowest score for the University students.

- b Compare the medians of both groups of students.

- c For which student group(s) is the mean greater than the median?

University Students		High School Students
7	0	
6 6 3	1	0 0 3 5
4 3 2 1	2	1 2 4 4 6
9 8 8 6	3	1 8 9
8 2	4	0 1
	5	
	6	
3	7	

Key: $4|1|2 = 14$ books and 12 books

13 The back to back stem and leaf plot shows the amount of cash (in dollars) carried by a random sample of teenage boys and girls:



- Which group carried more cash?
- Find the median amount of cash that the boys carried
- Find the median amount of cash that the girls carried.
- Which group's distribution is roughly bell shaped?
- Which group has more variation in the amounts of cash?
- Were there any outliers in the boys' amounts? If so, what are the value(s)?
- Were there any outliers in the girls' amounts? If so, what are the value(s)?

Boys		Girls
7	0	
1	1	1
5 4 1	2	2 6 8
8 5 4	3	3 4 4 6 6 8 9
9 8 2 2 2 1	4	3 4 6
9 7 4 3	5	4
8 5 2	6	
3 1	7	

Key: 1|2|2 = \$21 and \$22

14 The following back to back stem and leaf plot shows the length (in minutes) of a random sample of phone calls made by Sharon and Tricia:

- Who made a 14 minute phone call?
- Who has the higher median?
- Is Sharon's mean greater than her median?
- Is Tricia's mean greater than her median?

Sharon		Tricia
3	1	3 4
7 6 4 3 2	2	6 7 8
9 8	3	2 4
4 3	4	1 2
7 6	5	6 7 8

Key: 2|2|6 = 22 and 26

15 The back to back stem and leaf plot shows the number of pieces of paper used over several days by Charlie's and Dylan's students:

- Did Charlie's students use 7 pieces of paper on any day?
- Who's class had the higher median?
- Is the median greater than the mean in both groups?

Charlie's students		Dylan's students
7	0	7
3 2 1	1	3
8	2	8
4 3 2	3	3 4
9	4	5 6 7
2	5	2 3

Key: 1|1|3 = 11 and 13

16 The back to back stem and leaf plot shows the number of desserts ordered at Hotel A and Hotel B over several randomly chosen days:

- a Interpret the lowest score for Hotel A.
- b Which hotel's median is higher?
- c Is the mean greater than the median in both groups?

Hotel A		Hotel B
3	0	
4 3 2	1	3 4
7 6	2	7
4 3	3	3 4
6	4	6 7
2	5	2 3 4

Key: 2|1|3 = 12 and 13

17 The median house price in Oklachusetts is \$750 000 with a mean price of \$800 000 and the median house price in Brockway is \$750 000 with a mean price of \$680 000.

Which town is most likely to have some very expensive houses? Explain your answer.

18 Two Chemistry classes, each with 15 students, sit a 10-question multiple choice test, each with four possible answers (only one of which is correct). Their class results, out of 10, are below:

- Class 1: 3, 2, 1, 1, 2, 5, 2, 2, 2, 3, 2, 1, 5, 5, 4
- Class 2: 10, 6, 9, 8, 10, 9, 7, 7, 6, 8, 8, 10, 8, 9, 8

- a Complete the following table, rounding your answers to one decimal place when necessary:

	Mean	Median	Mode	Range
Class 1				
Class 2				

- b Which class was more likely to have studied for their test? Explain your answer.

19 The hours of sleep per night for two people over a two week period are shown below:



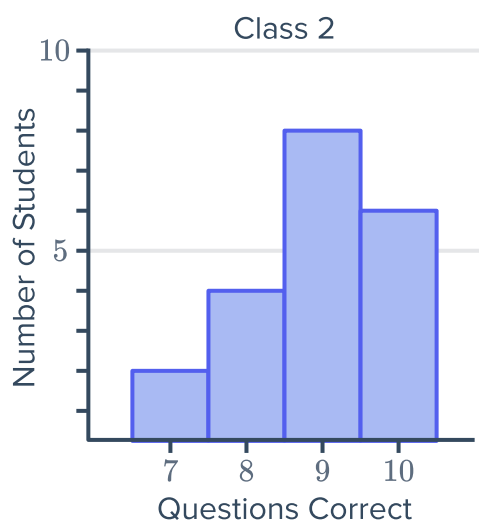
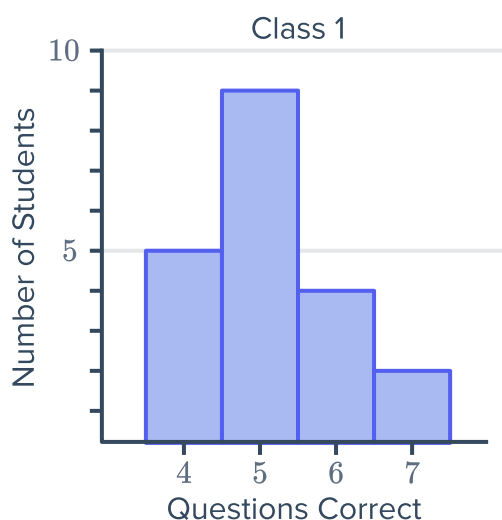
- Person A: 10, 9, 6, 6, 7, 8, 5, 6, 9, 9, 7, 5, 10, 9
- Person B: 7.5, 7.5, 8, 7.5, 7, 8, 7.5, 7, 7, 7, 7, 7, 8, 8

a Complete the following table, rounding your answers to one decimal place when necessary:

	Mean	Median	Mode	Range
Person A				
Person B				

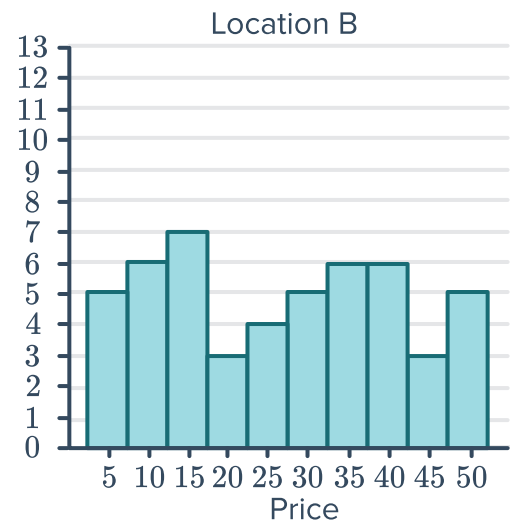
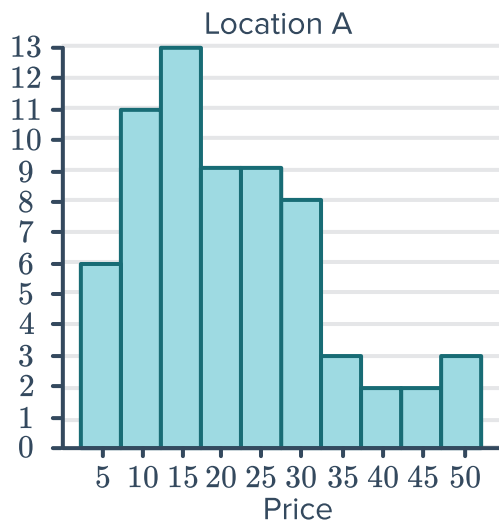
- b Which person seems to be the least consistent in their sleep habits?
- c Which statistical piece of evidence supports your answer in part (b)?
- d Which person had the most sleep over the 14 nights?
- e Which statistical piece of evidence supports your answer in part (d)?

20 Two Science classes, each with 20 students, were given a 10 question True/False test. The results for each class are shown below:



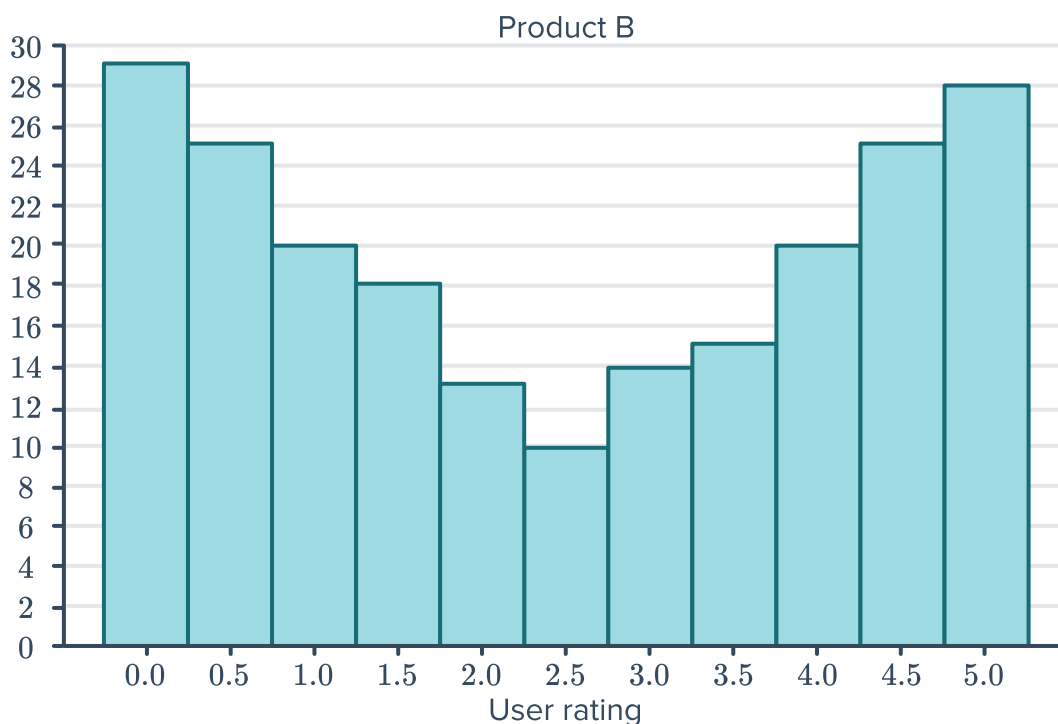
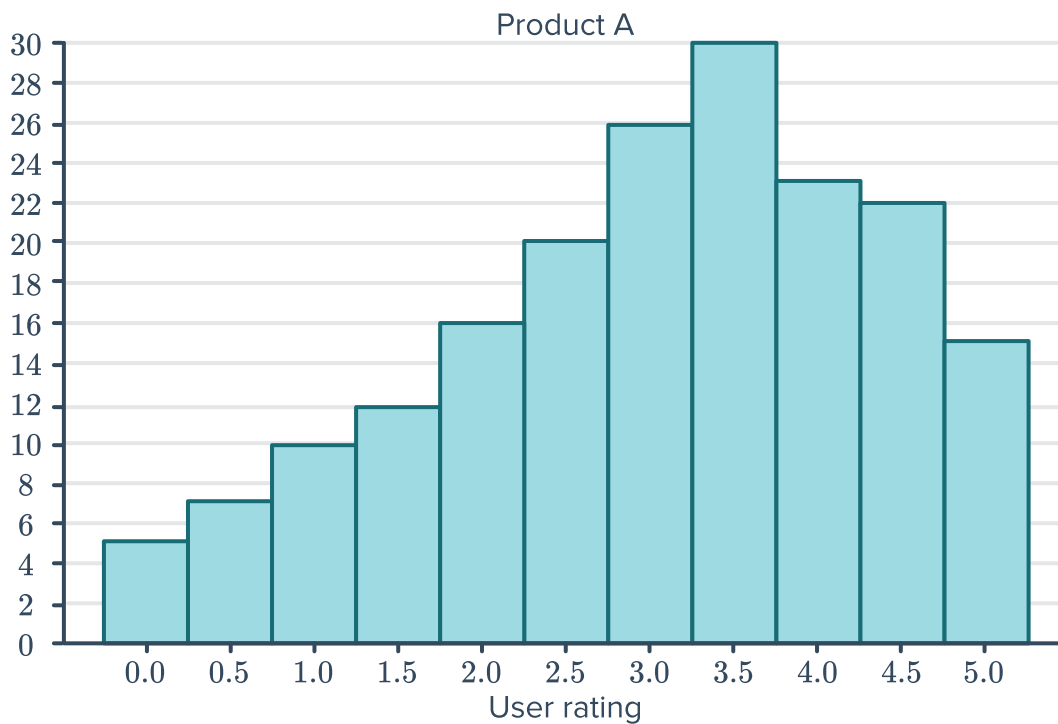
- a Calculate the mean for Class 1 correct to one decimal place.
- b Calculate the mean for Class 2 correct to one decimal place.
- c Find the range for Class 1.
- d Find the range for Class 2.
- e Do you think Class 1 studied for their test? Justify your answer.
- f Do you think Class 2 studied for their test? Justify your answer.

23 The prices of sales at two locations of a shop are recorded in the histograms below:



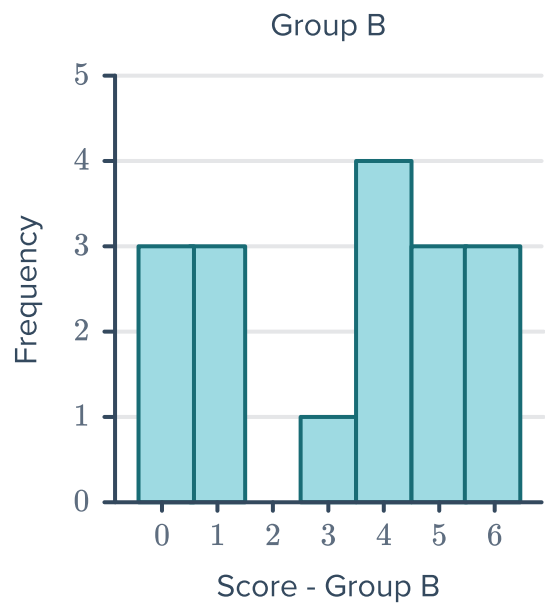
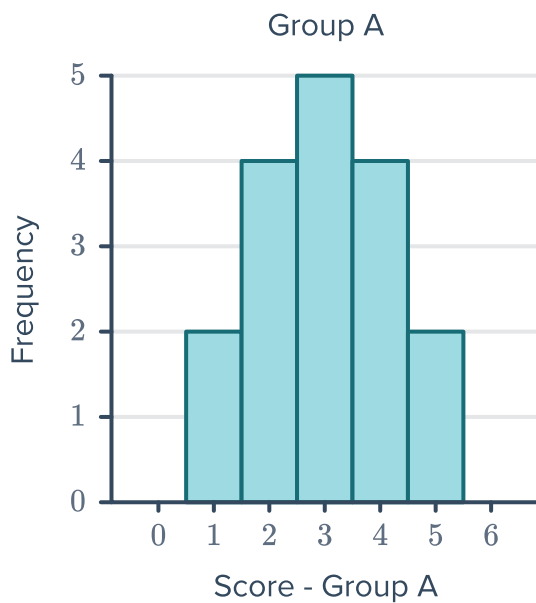
- a Find the mean sale price for:
- i Location A
 - ii Location B
- b Which location made more sales?
- c Which location had a higher average sale price?

24 An online shopping website records the user ratings for two similar products in the histograms below:



- a Find the median user rating for:
- i Product A
 - ii Product B
- b According to the website's rating system, a perfect rating is 5.0. Which product got more perfect ratings?
- c According to the website's rating system, a positive rating is greater than 3. Which product got more positive ratings?

25 The following histograms show the season results of two soccer groups, Group A and Group B, and the number of games (frequency) in which they scored a certain number of goals:



- a Find the mode for Group A.
- b Find the mode for Group B.
- c Find the range for Group A.
- d Find the range for Group B.
- e Which group scored the lowest total number of goals during the season?
- f Which group has the most varied results?